

Lab 1

Part 1: Capturing HTTP Traffic

The image shows a Wireshark network traffic capture window. The top pane displays a list of captured packets, with the first eight packets highlighted in green. The second pane shows the details of the selected packet (packet 8), which is an HTTP GET request for /online/. The third pane shows the raw packet data in hexadecimal and ASCII format.

No.	Time	Source	Destination	Protocol	Length	Info
1	8.473811000	192.168.0.194	34.223.124.45	HTTP	490	GET / HTTP/1.1
2	11.843260800	34.223.124.45	192.168.0.194	HTTP	1027	HTTP/1.1 200 OK (text/html)
3	34.959845600	2001:16a4:299:6bc5	2600:1f13:37c:1400	HTTP	592	GET /online/ HTTP/1.1
4	35.198393300	2001:16a4:299:6bc5	2600:1f13:37c:1400	HTTP	645	HTTP/1.1 301 Moved Permanently (text/html)
5	35.201616100	2001:16a4:299:6bc5	2600:1f13:37c:1400	HTTP	593	GET /online/ HTTP/1.1
6	35.509587100	2001:16a4:299:6bc5	2600:1f13:37c:1400	HTTP	293	HTTP/1.1 200 OK (text/html)
7	35.584424400	2001:16a4:299:6bc5	2600:1f13:37c:1400	HTTP	496	GET /favicon.ico HTTP/1.1
8	35.818707900	2001:16a4:299:6bc5	2600:1f13:37c:1400	HTTP	490	HTTP/1.1 200 OK (PNG)

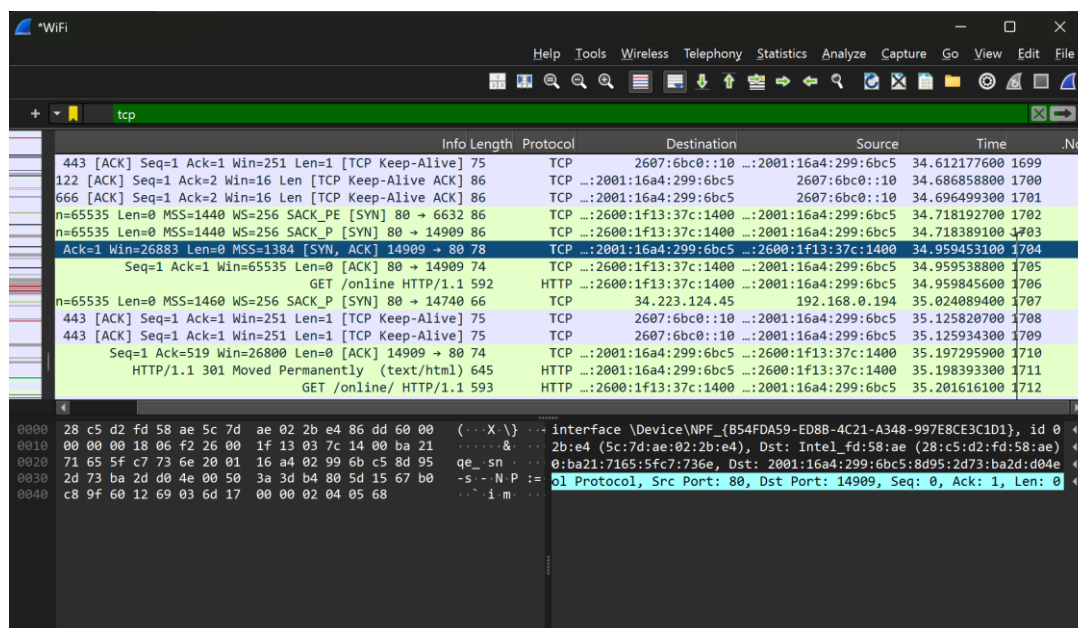
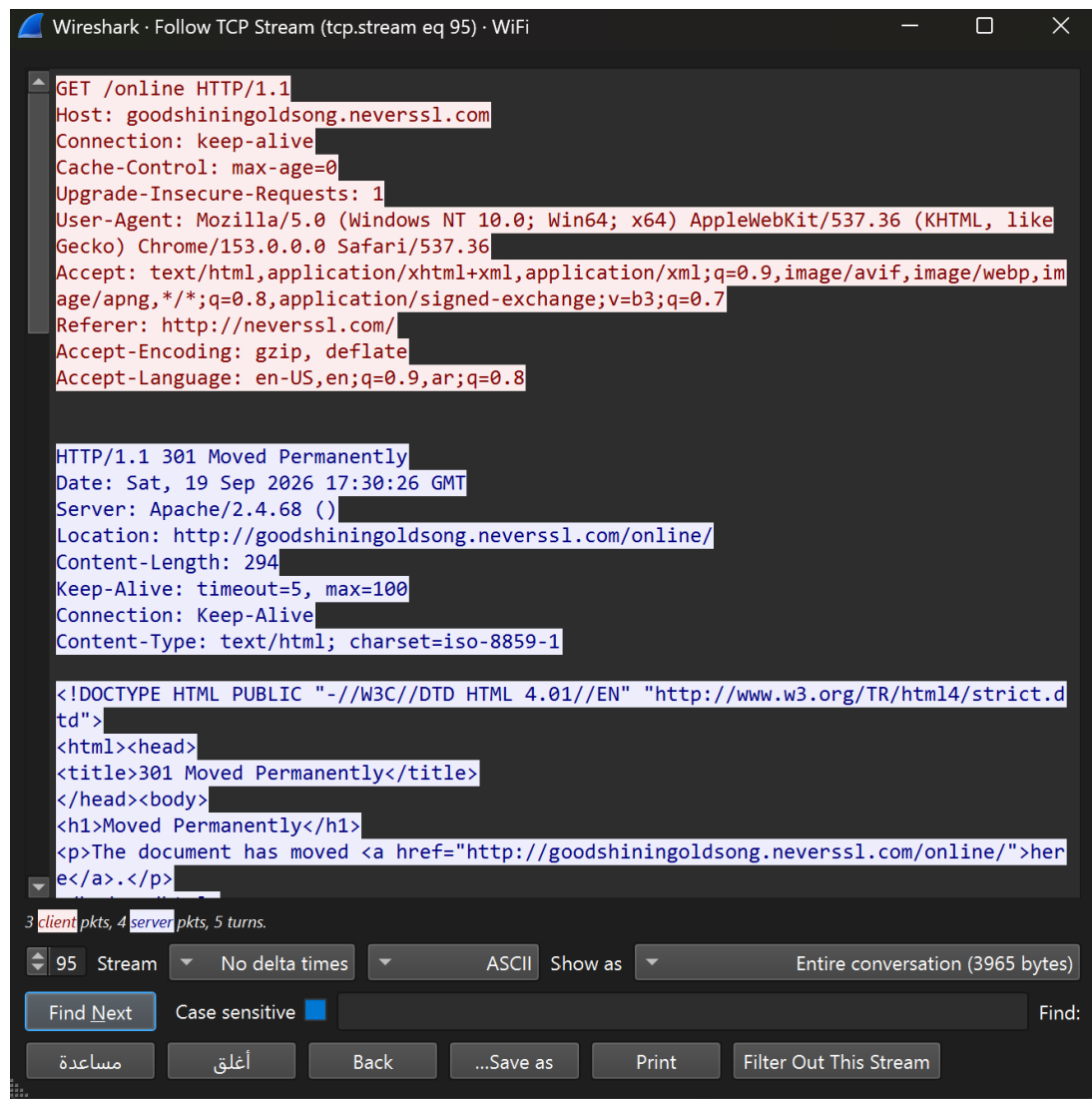
Details of packet 8 (HTTP GET /online/):

- Host: go
- Host: goldsong
- Host: l.com
- Host: n: keep-
- Host: ache-Con
- Host: x-age=0
- Host: -Insecu
- Host: ts: 1. U
- Host: t: Mozil
- Host: Windows
- Host: Win64

Raw packet data (hex):

```
0000 5c 7d ae 02 2b e4 28 c5 d2 fd 58 ae 86 dd 60 09
0010 a7 d9 02 1a 06 3f 20 01 16 a4 02 99 6b c5 8d 95
0020 2d 73 ba 2d d0 4e 26 00 1f 13 03 7c 14 00 ba 21
0030 71 65 5f c7 73 6e 3a 3d 00 50 67 b0 c8 9f b4 80
0040 5d 16 50 18 ff ff 47 f5 00 00 47 45 54 20 2f 6f
0050 6e 6c 69 6e 65 20 48 54 54 50 2f 31 2e 31 0d 0a
0060 48 6f 73 74 3a 20 67 6f 6f 64 73 68 69 6e 69 6e
0070 67 6f 6c 64 73 6f 6e 67 2e 6e 65 76 65 72 73 73
0080 6c 2e 63 6f 6d 0d 0a 43 6f 6e 6e 65 63 74 69 6f
0090 6e 3a 20 6b 65 65 70 2d 61 6c 69 76 65 0d 0a 43
00a0 61 63 68 65 2d 43 6f 6e 74 72 6f 6c 3a 20 6d 61
00b0 78 2d 61 67 65 3d 30 0d 0a 55 70 67 72 61 64 65
00c0 2d 49 6e 73 65 63 75 72 65 2d 52 65 71 75 65 73
00d0 74 73 3a 20 31 0d 0a 55 73 65 72 2d 41 67 65 6e
00e0 74 3a 20 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20 28
00f0 57 69 6e 64 6f 77 73 20 4e 54 20 31 30 2e 30 3b
0100 70 57 69 6e 36 3d 3b 20 78 36 3d 29 20 41 70 70
```

Part 2: Analyzing TCP/IP Traffic



Part 3: Capturing and Analyzing UDP Traffic

The image shows a Wireshark packet capture window titled "WiFi". The interface includes a menu bar (Help, Tools, Wireless, Telephony, Statistics, Analyze, Capture, Go, View, Edit, File) and a toolbar. A filter bar at the top shows "udp" selected. The packet list pane displays a series of DNS queries and responses for various domains, including default.exp-tas.com, api.telegram.org, and www.googleapis.com. The packet details pane shows the structure of a UDP packet, including the User Datagram Protocol (UDP) header and the data payload. The packet bytes pane shows the raw data in hexadecimal and ASCII format.

No.	Time	Source	Destination	Protocol	Length	Info
1337	24.010009800	192.168.0.1	192.168.0.194	DNS	202	query response 0x468b AAAA default.exp-tas.com CNAME
1384	26.101357800	192.168.0.194	192.168.0.1	DNS	76	Standard query 0x35bb AAAA api.telegram.org
1385	26.117381700	192.168.0.1	192.168.0.194	DNS	200	query response 0x35bb AAAA api.telegram.org AAAA
1557	30.757677600	192.168.0.194	192.168.0.1	DNS	78	Standard query 0x15c1 A www.googleapis.com
1558	30.757677600	192.168.0.194	192.168.0.1	DNS	78	Standard query 0xb119 AAAA www.googleapis.com
1559	30.772189000	192.168.0.1	192.168.0.194	DNS	206	query response 0x15c1 A www.googleapis.com A 172.217.206
1560	30.774864000	192.168.0.1	192.168.0.194	DNS	302	query response 0xb119 AAAA www.googleapis.com AAAA 2
1588	31.264883000	192.168.0.194	192.168.0.1	DNS	79	Standard query 0xc02d AAAA play.googleapis.com
1589	31.264911400	192.168.0.194	192.168.0.1	DNS	79	Standard query 0xb0ae A play.googleapis.com
1590	31.269507000	192.168.0.1	192.168.0.194	DNS	303	d query response 0xc02d AAAA play.googleapis.com AAAA
1591	31.269511000	192.168.0.1	192.168.0.194	DNS	207	query response 0xb0ae A play.googleapis.com A 172.21.207
1674	33.882627300	2001:16a4:299:6bc5	2001:16a4:299:6bc5	UDP	443	Len=29 443 → 63523 91
1679	33.994122700	2001:16a4:299:6bc5	2001:16a4:299:6bc5	UDP	87	Len=25 63523 → 443 87
1729	35.461284900	2001:16a4:299:6bc5	2001:16a4:299:6bc5	UDP	91	Len=29 443 → 63523 91

Packet 1679 details:

- Ethernet II, Src: Intel FD58:AE (28:c5:d2:fd:58:ae), Dst: Intel FD58:AE (28:c5:d2:fd:58:ae)
- Internet Protocol Version 4, Src: 2001:16a4:299:6bc5, Dst: 2001:16a4:299:6bc5
- User Datagram Protocol, Src Port: 443, Dst Port: 63523
- Data (25 bytes)

Packet 1679 bytes:

```
28 c5 d2 fd 58 ae 5c 7d ae 02 2b e4 86 dd 60 00 (...X\}... interface \Device\NPF_{B54FDA59-ED8B-4C21-A348-997E8CE3C1D1}, id 0
0010 00 00 00 21 11 37 20 01 48 60 48 02 00 32 00 00 ...!7... H\ 2b:e4 (5c:7d:ae:02:2b:e4), Dst: Intel_fd:58:ae (28:c5:d2:fd:58:ae)
0020 00 00 00 00 00 78 20 01 16 a4 02 99 6b c5 8d 95 ...X... 2001:4860:4802:32::78, Dst: 2001:16a4:299:6bc5:8d95:2d73:ba2d:d04e
0030 2d 73 ba 2d d0 4e 01 bb f8 23 00 21 fb 8d 5d 9f -s- N-...#
0040 26 08 51 3d a2 16 c8 ab 17 ee 73 49 a8 ee ac 4e & Q=...:
0050 aa 59 1b 34 c9 fe bf Y.4...
```

Part 4: Comparing TCP and UDP

	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	TCP creates a connection before sending data and provides reliable communication.
Data Integrity and Ordering	TCP	TCP checks errors and keeps data in the correct order.

Task 2: Identify the use Cases and Performance of TCP and UDP.

	TCP	UDP
Use cases	Websites, Email, File Transfer	Video Streaming, Online Games, Voice Calls
Performance	Slower but reliable	Faster but less reliable